

3 Three planes have the following equations.

$$2x - 3y + z = -3,$$

$$x - 4y + 2z = 1,$$

$$-3x - 2y + 3z = 14.$$

(a) (i) Write the system of equations in matrix form. **[1]**

(ii) Hence find the point of intersection of the planes. **[2]**

(b) In this question you must show detailed reasoning.

Find the acute angle between the planes $2x - 3y + z = -3$ and $x - 4y + 2z = 1$. **[4]**